01 · EXECUTIVE SUMMARY

# Procurement, re-engineered for the Group.

A multi-tenant, audit-grade procurement command center — replacing spreadsheet-driven workflows with a governed, integrated, scalable system of record.

**11**
Domain modules covering the full case lifecycle

**6**
Tenant-isolated database schemas with row-level security

**4**
Asynchronous queues
imports · exports · alerts · projections

**100%**
Mutating actions written to an append-only audit trail

### STRATEGIC OUTCOME

A single, governed source of truth for procurement across legal entities and departments.

### ENTERPRISE READINESS

Production-grade controls — RLS, RBAC, CSRF, Argon2id, idempotency, append-only audit.

### BUILT TO SCALE

Stateless tier, per-queue worker scale-out, managed-DB & Kubernetes path defined.

02 · BUSINESS PROBLEM & OPPORTUNITY

# From spreadsheet sprawl to a procurement system of record.

Procurement teams spend more time reconciling than deciding. ProcureDesk closes the governance, visibility, and accountability gaps that legacy tooling cannot.

## TODAY — OPERATING REALITY

- Procurement tracked across siloed spreadsheets
- No reliable audit trail; decisions hard to defend
- Manual reconciliation across entities and departments
- Contract & RC/PO expiries surfaced too late
- Cross-entity governance brittle and informal

## WITH PROCUREDESK — TARGET STATE

- Single tenant-isolated source of truth
- Append-only audit on every state change
- Automated imports, exports, and notifications
- Forward-looking expiry projections & saved views
- Role- and entity-scoped access by design

03 · SOLUTION ARCHITECTURE

# Modular, asynchronous, multi-tenant by design.

Four cleanly separated layers — designed to scale horizontally and isolate tenants at the data tier.

| | |
|---|---|
| EXPERIENCE | React 19 SPA · TanStack Router & Query · role-aware UI |
| EDGE | Nginx · TLS termination · rate limiting · CSRF · WAF-ready |
| APPLICATION | NestJS modular monolith (11 domains) · BullMQ worker · transactional outbox |
| DATA & INTEGRATION | PostgreSQL 16 with RLS · Redis 7 · Azure Blob · Microsoft Graph · Prometheus + Grafana |

## ARCHITECTURAL PRINCIPLES

A **Stateless application tier.** Sessions in DB, counters in Redis — linear horizontal scale.

B **Transactional outbox.** At-least-once async dispatch without distributed transactions.

C **Defence in depth.** Service-layer checks plus Postgres row-level security as the floor.

D **Forward-only schema.** Backward-compatible migrations enable safe rolling deploys.

04 · CORE CAPABILITIES

# One platform — the full procurement lifecycle.

Six capability domains, an asynchronous backbone, and one continuous workflow from planning to renewal.

### ID Identity & Access

Multi-tenant RBAC, entity scopes, password policy, session lifecycle.

### CT Catalog & Reference

Tender types, stage policies, completion rules, tenant choice categories.

### PC Procurement Cases

Lifecycle, milestones, financials, delays, governed transitions.

### AP Awards & Planning

Case awards, RC/PO plans, tender-plan-to-case linkage.

### RP Reporting

Case facts, contract-expiry projections, saved views, dashboards.

### AS Async Backbone

Excel imports/exports, notifications, audit trail, outbox & DLQ.

STAGE 1 **Plan** → STAGE 2 **Case** → STAGE 3 **Approve** → STAGE 4 **Award** → STAGE 5 **Track** → STAGE 6 **Renew**

05 · SECURITY · SCALABILITY · INTEGRATIONS

# Built for trust, engineered for scale.

Enterprise controls baked in from day one, with a defined path from single-host launch to group-wide multi-tenant scale.

### S GOVERNANCE & SECURITY

- PostgreSQL row-level security per tenant
- RBAC + entity scopes + access levels
- Argon2id, CSRF, signed sessions, login throttle
- Append-only audit · idempotency keys
- CI gates: Gitleaks · pnpm audit · Trivy CVE

### ↗ SCALABILITY

- Stateless API — linear horizontal scale
- Per-queue worker autoscaling
- Read-replica + PgBouncer ready
- Kubernetes target topology defined
- Edge CDN / WAF path identified

### ⬡ INTEGRATIONS

- Microsoft Graph — notification email
- Azure Blob — private file storage
- Prometheus / Grafana — observability
- GitHub Actions / GHCR — secure delivery
- SSO / OIDC — extension seam

06 · BUSINESS IMPACT & ROADMAP

# Measurable impact, a clear path to group-wide scale.

From production handover today, to multi-tenant rollout, Kubernetes-native operations, and AI-assisted procurement insight.

**AUDIT POSTURE**

**7 yrs+**

Append-only retention supports SOX-aligned governance.

**ASYNC LATENCY**

**< 10 s**

End-to-end outbox dispatch under target SLO.

**RELEASE SAFETY**

**N-1**

Forward-only, backward-compatible migrations.

**BULK INGEST**

**25 MiB**

Per Excel job, with per-row reconciliation.

## PHASED ROADMAP

**NOW**
Production handover · single-tenant launch

**Q + 1**
Multi-tenant rollout · managed Postgres + replica

**Q + 2**
Kubernetes HPA · per-queue workers · edge CDN / WAF

**Q + 3**
Supplier portal · e-tendering capabilities

**Q + 4**
AI-assisted catalog & case insights